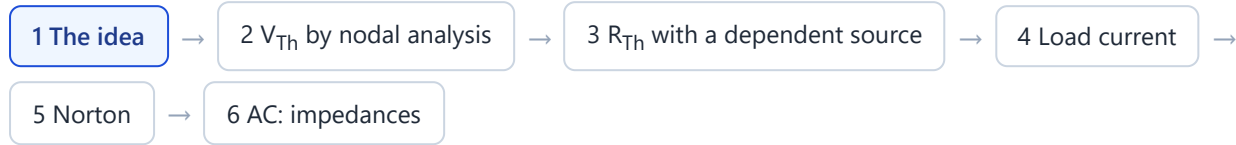


06 — Thévenin's and Norton's theorems

Class notes p.10–11, p.15, set 2 p.10–13 · MTE syllabus item 5 (DC and AC) · 1 board question (with a dependent source) + 1 textbook AC problem · needs 01, 03, 04, 05

Map



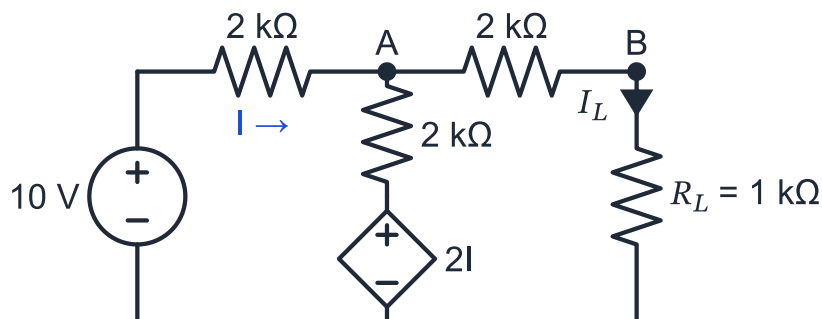
The question this file answers

- **Class p.11 · set 2 p.11** Determine the load current using Thévenin's theorem (10 V source, dependent source $2I$, $R_L = 1\text{ k}\Omega$).
- **textbook** AC: $10\angle 0^\circ\text{ V}$, $1\text{ k}\Omega$, $1\text{ }\mu\text{F}$ at $\omega = 1000\text{ rad/s}$. Find the Thévenin and Norton equivalents.

Build

CLASS P.11 · SET 2 P.11

Determine the load current I_L using Thévenin's theorem.



The board circuit. I is the current in the left $2\text{ k}\Omega$, flowing right. The dependent source is a CCVS of value $2I$, with $+$ at the top.

UNITS

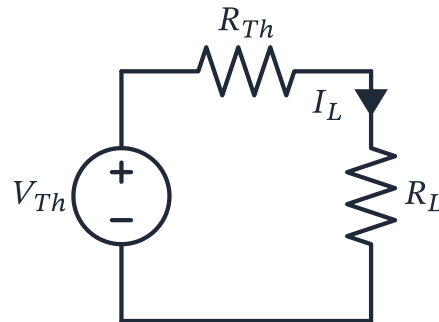
With $\text{k}\Omega$ and mA , " $2I$ " means $2\text{ k}\Omega \times I$ (01, step 3): $I = 1\text{ mA}$ gives 2 V . The set 2 working reads it differently and gets $V_{\text{Th}} = 5.0025\text{ V}$; the Note at the end of step 2 explains why both are right for their reading. The rest below is the standard method, with every number confirmed by computer.

GUESS FIRST

Without solving: if R_L changed from $1\text{ k}\Omega$ to $5\text{ k}\Omega$, would you want to re-solve the whole circuit? What would you rather have?

1 • The idea

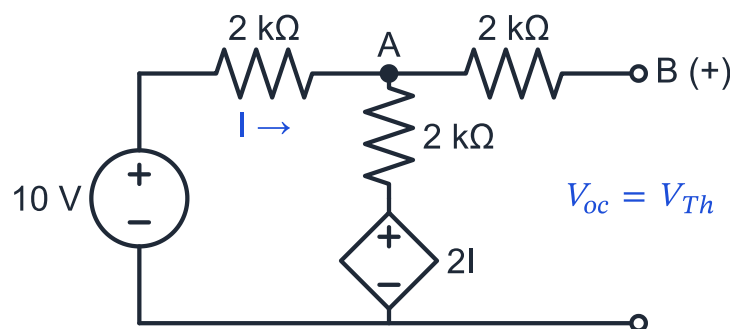
Any **linear** two-terminal network, however complicated, behaves at its terminals **exactly like one voltage source V_{Th} in series with one resistance R_{Th}** . Replace the whole network with that pair and the load is easy, for any R_L .



The Thévenin equivalent: $I_L = V_{Th} / (R_{Th} + R_L)$.

Procedure (class p.10): 1. Remove the load R_L (or Z_L). 2. Find V_{Th} , the **open-circuit** voltage across the load terminals. 3. Find R_{Th} across the same terminals. 4. Put R_L back on the equivalent: $I_L = V_{Th} / (R_{Th} + R_L)$.

2 • V_{Th} by nodal analysis (the professor's route)



Load removed. No current flows in the right $2\text{ k}\Omega$ (it leads to an open end), so $V_{Th} = V_A$.

This is nodal analysis (04) with one unknown, V_A . Units: V, mA, $\text{k}\Omega$.

- Current in the left resistor, into A: $I = \frac{10 - V_A}{2}$
- Current in the middle branch, down: $I_1 = \frac{V_A - 2I}{2}$. No current in the right $2\text{ k}\Omega$, so KCL at A gives $I = I_1$ (set 2 p.11 writes exactly this).
- KCL: $\frac{10 - V_A}{2} = \frac{V_A - 2I}{2} \Rightarrow 10 - V_A = V_A - 2I$
- Substitute the controlling current, $2I = 2 \times \frac{10 - V_A}{2} = 10 - V_A$:

$$10 - V_A = V_A - (10 - V_A) \quad \Rightarrow \quad V_A = \frac{20}{3} \text{ V} \quad (\text{and } I = \frac{5}{3} \text{ mA})$$

$$V_{Th} = 20/3 \approx 6.67 \text{ V}$$

CHECK 1

Why does no current flow in the right-hand 2 k Ω here, and why does that make $V_{Th} = V_A$?

SET 2 P.12 GETS $V_{Th} = 5.0025 \text{ V}$. NOT AN ARITHMETIC ERROR: A UNIT READING

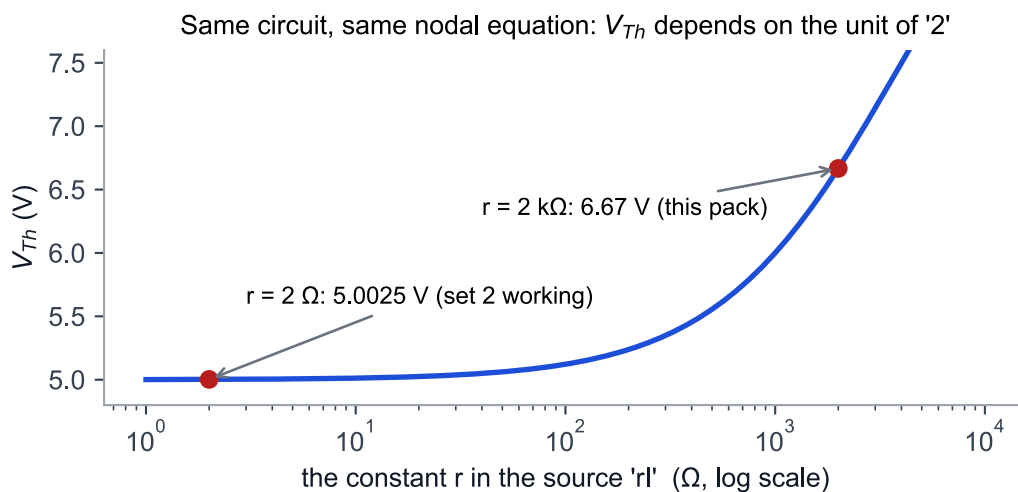
The set 2 working writes $I = \frac{10 - V_A}{2k}$ **in amperes** ($2k = 2000 \Omega$) and then uses the source as $2 \times I$ volts.

That treats the constant as **2 Ω** :

$$10 - V_A = V_A - 2 \cdot \frac{10 - V_A}{2000} \quad \Rightarrow \quad 10.01 = 2.001 V_A \quad \Rightarrow \quad V_A = 5.0025 \text{ V}$$

For that reading the arithmetic is right: a computer solve with a 2 Ω source gives 5.0025 V, $R_{Th} = 2.9995 \text{ k}\Omega$, $I_L = 1.2508 \text{ mA}$.

The pack keeps **2 k Ω** ($V_{Th} = 20/3 \text{ V}$, $I_L = 20/11 \text{ mA}$) because in a k Ω circuit written in mA, "2I" normally means 2 V per mA. And a 2 Ω source against 2 k Ω resistors changes V_{Th} by only 0.0025 V from the 5 V it would be without the source, so the question could not be showing off a dependent source. **Ask the professor which unit the "2" has.** In the exam, state your reading in one line ("2I with I in mA, i.e. 2 k Ω ") before solving.



V_{Th} from the same nodal equation for every value of the constant. The two readings are two points on one curve.

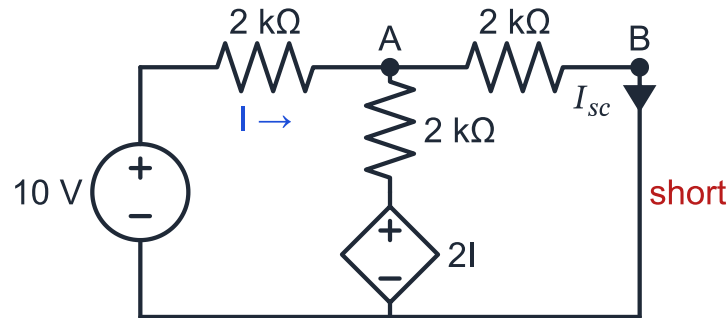
3 · R_{Th} when there is a dependent source

GUESS FIRST

Switch off the 10 V source and "look in" from B: 2 k Ω in series with ($2 \text{ k}\Omega \parallel 2 \text{ k}\Omega$) = 3 k Ω . What is wrong with doing that here?

The dependent source **stays on** (05), and it changes how current flows. So resistors alone don't give R_{Th} . Two correct methods; both give the same answer.

Method A: $R_{Th} = V_{oc} / I_{sc}$. Short the terminals and find the short-circuit current:

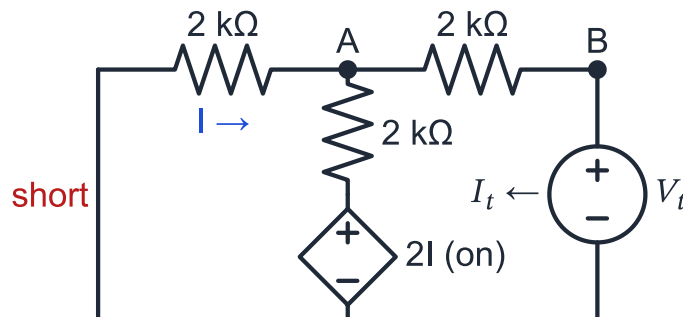


B shorted to the bottom rail. Now the right $2\text{ k}\Omega$ carries current too.

KCL at A: $I = \frac{V_A - 2I}{2} + \frac{V_A}{2} \Rightarrow V_A = 2I$. With $I = \frac{10 - 2I}{2}$: $I = 2.5\text{ mA}$, $V_A = 5\text{ V}$, so $I_{sc} = \frac{5}{2} = \mathbf{2.5\text{ mA}}$.

$$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{20/3}{2.5} = 8/3 \approx 2.67\text{ k}\Omega$$

Method B: test source (the professor's method, class p.10: $R_{Th} = V_{dc} / I_{dc}$). Switch off the **independent** source only, keep $2I$, apply a test voltage V_t at the terminals and find the current I_t it drives:



10 V replaced by a short; the dependent source is still on; the test source V_t is at B.

With the 10 V shorted, the left $2\text{ k}\Omega$ runs from ground to A, so $I = \frac{0 - V_A}{2} = -\frac{V_A}{2}$. Currents leaving A downwards:

- through the left $2\text{ k}\Omega$: $\frac{V_A}{2}$
- through the middle branch: $\frac{V_A - 2I}{2} = \frac{V_A + V_A}{2} = V_A$

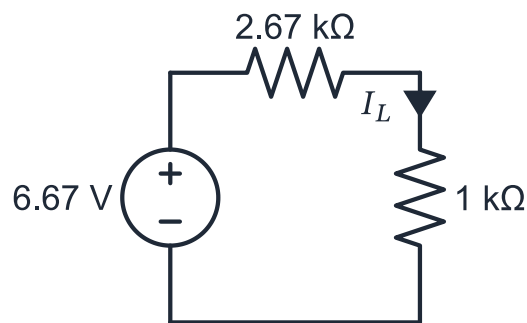
Total = $1.5 V_A$, so A looks like $\frac{1}{1.5} = \frac{2}{3}\text{ k}\Omega$ to ground. Then $R_{Th} = 2 + \frac{2}{3} = \mathbf{8/3\text{ k}\Omega}$ ✓, the same as method A.

The 3 kΩ from the Guess box is **wrong**: it ignores that the dependent source doubles the middle branch's current.

CHECK 2

(a) In method B, why is I negative? (b) If the circuit had **no** independent source at all, which method would still work, and why?

4 • The load current



The whole left side, replaced by its Thévenin equivalent.

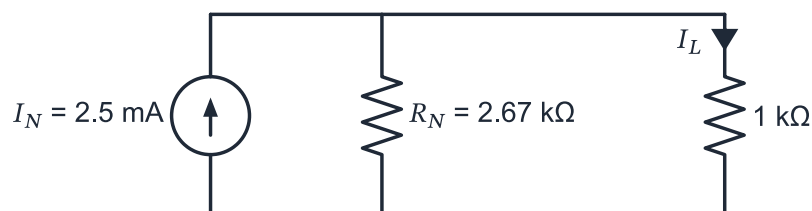
$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{20/3}{8/3 + 1} = \frac{20}{11} \approx 1.82 \text{ mA}$$

Confirmed by solving the full original circuit with R_L in place: **1.818 mA**. For $R_L = 5 \text{ k}\Omega$ it is now one line: $6.67 / 7.67 = 0.87 \text{ mA}$.

5 • Norton: the same network as a current source

Norton's equivalent is a **current source I_N in parallel with R_N** (class p.15). By source transformation (03, step 4):

$$I_N = I_{sc} = \frac{V_{Th}}{R_{Th}} \quad R_N = R_{Th}$$



Norton equivalent of the same network: 2.5 mA in parallel with 2.67 kΩ.

Current division: $I_L = 2.5 \times \frac{8/3}{8/3 + 1} = \mathbf{20/11 \text{ mA}}$ ✓. Same answer.

Norton procedure (class p.15): (i) remove the load; (ii) find R_N , exactly as for R_{Th} ; (iii) short the load terminals and find the short-circuit current I_N ; (iv) draw $I_N \parallel R_N$ with the load.

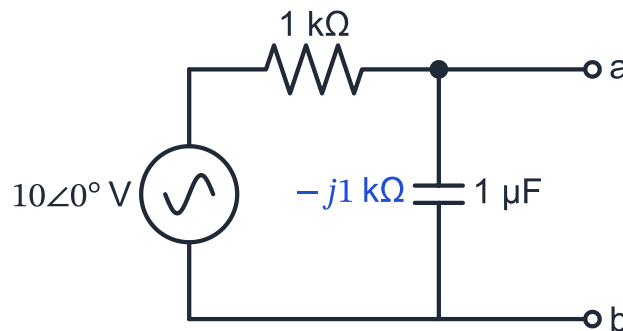
CHECK 3

A network has $V_{Th} = 12 \text{ V}$ and $R_{Th} = 4 \text{ k}\Omega$. Give its Norton equivalent.

6 • Thévenin and Norton with AC sources

TEXTBOOK

A source $10\angle 0^\circ \text{ V}$ (peak) at $\omega = 1000 \text{ rad/s}$ drives $1 \text{ k}\Omega$ in series, with $1 \mu\text{F}$ across the output a–b. Find the Thévenin and Norton equivalents at a–b.



$$\text{The capacitor's impedance at } \omega = 1000: Z_C = \frac{1}{j\omega C} = \frac{1}{j \cdot 1000 \cdot 10^{-6}} = -j1000 \Omega.$$

GUESS FIRST

Is the open-circuit voltage 10 V, less, or more? Is it in phase with the source?

Nothing new in the method: the same four steps, with every resistance replaced by an **impedance** Z ($R \rightarrow R$, $L \rightarrow j\omega L$, $C \rightarrow \frac{1}{j\omega C}$) and every value a **phasor** (a complex number carrying amplitude and phase). Work in $\text{k}\Omega$:

$$Z_C = -j1 \text{ k}\Omega.$$

V_{Th} : open circuit, so the capacitor and the $1 \text{ k}\Omega$ form a voltage divider (02):

$$V_{Th} = 10 \cdot \frac{-j1}{1 - j1} = 10 \cdot \frac{-j(1 + j)}{2} = 5 - j5 = 7.07\angle -45^\circ \text{ V}$$

Z_{Th} : source $\rightarrow$ short, look in from a–b: $1 \text{ k}\Omega$ in parallel with $-j1 \text{ k}\Omega$:

$$Z_{Th} = \frac{1 \cdot (-j1)}{1 - j1} = 0.5 - j0.5 \text{ k}\Omega = 500 - j500 \Omega$$

Norton: $I_N = \frac{V_{Th}}{Z_{Th}} = \frac{5 - j5}{0.5 - j0.5} = 10\angle 0^\circ \text{ mA}$. Check: shorting a–b shorts the capacitor, leaving 10 V across 1 k Ω : 10 mA ✓.

The Guess answer: **less** (7.07 V) and **lagging** by 45°. Unlike the resistor divider in 02, an impedance divider changes the phase.

The Thévenin equivalent is $7.07\angle -45^\circ \text{ V}$ in series with $500 - j500 \Omega$ (a resistor with a capacitive part). 07 §4 finds the load that takes the most power from it.

CHECK 4

Same circuit at $\omega = 2000 \text{ rad/s}$. Find Z_C and V_{Th} .

Exam form

Thévenin's theorem. Any linear bilateral network seen from two terminals can be replaced by a voltage source V_{Th} (the open-circuit voltage at those terminals) in series with R_{Th} (the resistance seen at the terminals with all independent sources switched off).

Norton's theorem. The same network can be replaced by a current source I_N (the short-circuit current) in parallel with $R_N = R_{Th}$.

Link: $V_{Th} = I_N \cdot R_{Th}$.

R_{Th} with dependent sources: keep them on; use V_{oc}/I_{sc} or apply a test source: $R_{Th} = V_t/I_t$.

AC: the same, with impedances ($Z_L = j\omega L$, $Z_C = 1/j\omega C$) and phasors: V_{Th} and Z_{Th} are complex; $I_N = V_{Th}/Z_{Th}$.

Traps

- Switching off the dependent source when finding R_{Th} .
- Using the test-source method but forgetting to switch off the **independent** sources (the class p.10 sketch still shows V_{in} ; it must be zeroed).
- Measuring V_{Th} with the load still connected.
- Norton: I_N is the **short-circuit** current, not the current in the original load.
- Mixing A and mA in one nodal equation (set 2 p.12): decide the unit of every dependent-source constant first.
- AC: adding impedance magnitudes instead of the complex numbers (1 k Ω and $-j1 \text{ k}\Omega$ in series is 1.414 k Ω in size, not 2 k Ω).

Self-test

1. State Thévenin's theorem in two lines.
2. Which two measurements, both at the terminals, give V_{Th} and R_{Th} for any linear network?
3. For the board circuit, what would R_{Th} be if the dependent source were replaced by a plain wire?
4. Find I_L for the board circuit if $R_L = 8/3 \text{ k}\Omega$. What is special about that value? (Hint: 07.)
5. Convert $V_{Th} = 6 \text{ V}$, $R_{Th} = 3 \text{ k}\Omega$ into Norton form.

Answers

Check 1. The right $2\text{ k}\Omega$ ends at an open terminal, so there is no path: $I = 0$, voltage drop = 0, so $V_B = V_A$.

Check 2. (a) With the 10 V shorted, current flows from ground *into* A through the left resistor, which is opposite to the defined direction of I . (b) Only the test-source method: $V_{oc} = 0$ and $I_{sc} = 0$, so V_{oc}/I_{sc} is $0/0$.

Check 3. $I_N = 12/4 = \mathbf{3\text{ mA}}$ in parallel with $\mathbf{4\text{ k}\Omega}$.

Check 4. $Z_C = 1/(j \cdot 2000 \cdot 10^{-6}) = \mathbf{-j500\ \Omega}$. $V_{Th} = 10 \times \frac{-j0.5}{1 - j0.5} = \mathbf{4.47\angle -63.4^\circ\text{ V}}$ ($= 2 - j4\text{ V}$).

Self-test.

1. See Exam form.
2. The open-circuit voltage ($= V_{Th}$) and the short-circuit current (I_{sc}); $R_{Th} = V_{oc}/I_{sc}$.
3. The middle branch becomes a plain $2\text{ k}\Omega$: $R_{Th} = 2 + (2 \parallel 2) = \mathbf{3\text{ k}\Omega}$.
4. $\frac{20/3}{16/3} = \mathbf{1.25\text{ mA}}$. $R_L = R_{Th}$: this load receives the maximum possible power.
5. $\mathbf{2\text{ mA}}$ in parallel with $\mathbf{3\text{ k}\Omega}$.